

CH 12 HW #1

Due: 11:59pm on Friday, August 28, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

Problem 12.16

A particle is moving along a straight line with an initial velocity of 9 m/s when it is subjected to a deceleration of $a = (-1.2 v^{1/2}) \text{ m/s}^2$, where v is in m/s .

Part A

Determine how far it travels before it stops.

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$s =$

Part B

How much time does this take?

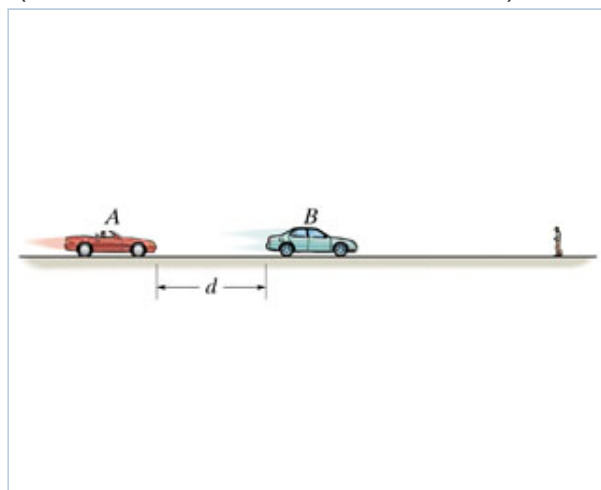
Express your answer to three significant figures and include the appropriate units.

ANSWER:

$t =$

Problem 12.17

Car B is traveling a distance d ahead of car A . Both cars are traveling at 60 ft/s when the driver of B suddenly applies the brakes, causing his car to decelerate at 11 ft/s^2 . It takes the driver of car A 0.75 s to react (this is the normal reaction time for drivers). When he applies his brakes, he decelerates at 16 ft/s^2 .



Part A

Determine the minimum distance d between the cars so as to avoid a collision.

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$$d = \text{[input box]}$$

Problem 12.23

A particle is moving along a straight line such that its acceleration is defined as $a = (-2v) \text{ m/s}^2$, where v is in meters per second. Suppose that $v = 20 \text{ m/s}$ when $s = 0$ and $t = 0$.

Part A

Determine the particle's position as function of time.

Express your answer in terms of t , where t is in seconds.

ANSWER:

$$s = \text{[input box]} \text{ m}$$

Part B

Determine the particle's velocity as function of time.

Express your answer in terms of t , where t is in seconds.

ANSWER:

$$v = \text{[input box]} \text{ m/s}$$

Part C

Determine the particle's acceleration as function of time.

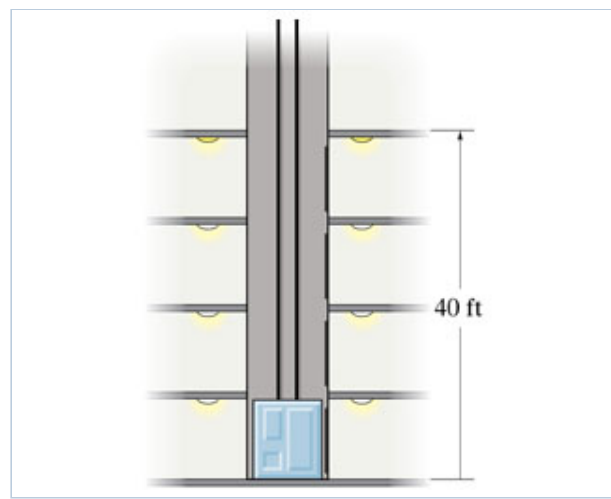
Express your answer in terms of t , where t is in seconds.

ANSWER:

$$a = \text{[input box]} \text{ m/s}^2$$

Problem 12.41

The elevator starts from rest at the first floor of the building. It can accelerate at 5 ft/s^2 and then decelerate at 2 ft/s^2 .



Part A

Determine the shortest time it takes to reach a floor 40 ft above the ground. The elevator starts from rest and then stops.

Express your answer using three significant figures and include the appropriate units.

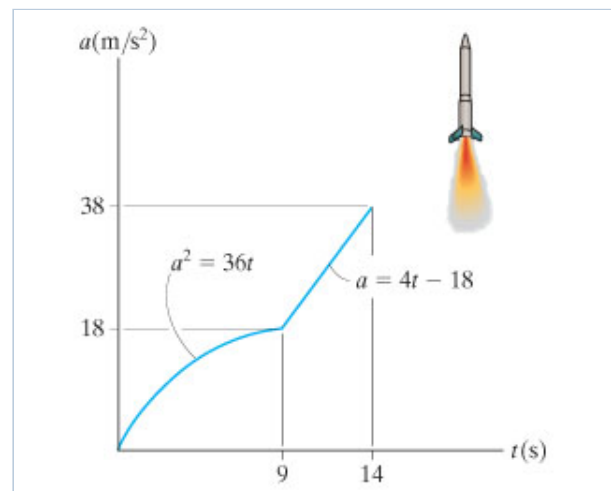
ANSWER:

$t =$

Part B Complete previous part(s)

Problem 12.47

The rocket has an acceleration described by the graph in .



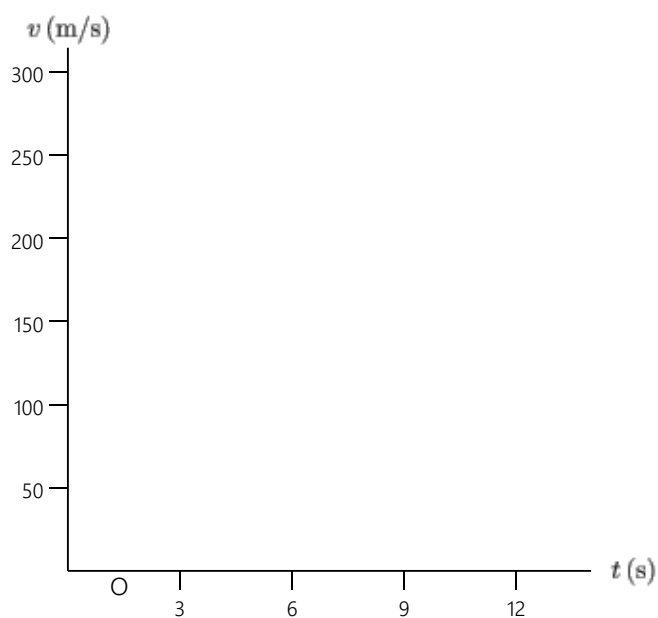
Part A

If it starts from rest, construct the v - t graph for the motion for the time interval $0 \leq t \leq 14$ s.

ANSWER:



No elements selected



Select the elements from the list and add them to the canvas setting the appropriate attributes. Press **TAB** to get to the main menu.

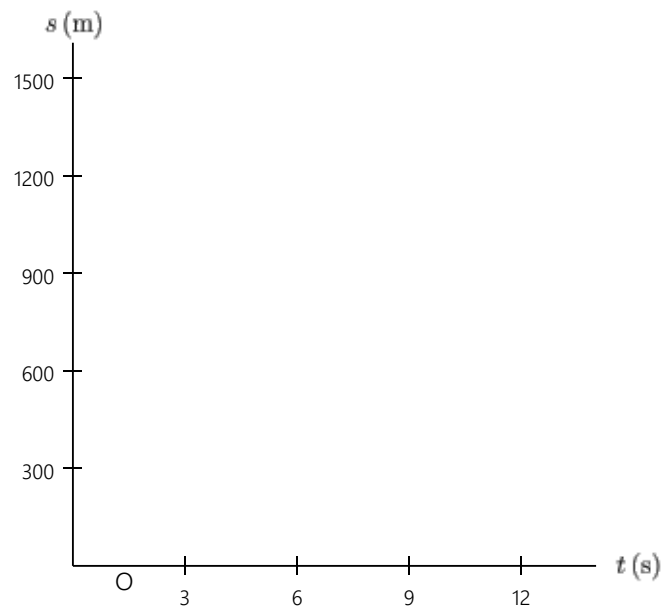
Part B

Construct the s - t graph for the motion for the time interval $0 \leq t \leq 14$ s.

ANSWER:



No elements selected



Select the elements from the list and add them to the canvas setting the appropriate attributes. Press **TAB** to get to the main menu.

Score Summary:

Your score on this assignment is 0.0%.

You received 0 out of a possible total of 5 points.